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Studierichting: Informatica
Oefeningenreeks 2D nr. 6.1

$$f(x) = \frac{x^2 - x - 6}{x^2 - 2x - 3} = \frac{(x-3)(x+2)}{(x-3)(x+1)} = \frac{x+2}{x+1} \text{ voor } x \neq 3$$

- $\lim_{x \to -1} f(x)$

  $$\lim_{x \to -1+} \frac{x+2}{x+1} = +\infty$$

  $$\lim_{x \to -1-} \frac{x+2}{x+1} = -\infty$$

- $\lim_{x \to -2} f(x)$

  $$\lim_{x \to -2} \frac{x+2}{x+1} = 0$$

- $\lim_{x \to 4} f(x)$

  $$\lim_{x \to 4} \frac{x+2}{x+1} = \frac{4+2}{4+1} = \frac{6}{5}$$

- $\lim_{x \to 3} f(x)$

  $$\lim_{x \to 3} \frac{x+2}{x+1} = \frac{3+2}{3+1} = \frac{5}{4}$$

- $\lim_{x \to \infty} f(x)$

  $$\lim_{x \to \infty} \frac{x+2}{x+1} = 1$$

References